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Smart Agriculture Advisor for Lebanese Farmers

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01

Description

Description

Smart Agriculture Advisor for Lebanese Farmers system answers agriculture questions across crop guidance, pests, irrigation, soil, fertilizer, and market timing, and it combines RAG, multi-agent orchestration, MCP tools, FastAPI, and Dockerized microservices.

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What I **Built**

What I Built

- Agent System A: LangGraph supervisor + specialists
- Agent System B: independent irrigation/planning service
- MCP server: soil, fertilizer, water-use tools
- Qdrant: vector database for RAG
- FastAPI + Docker Compose: full microservice deployment
- Frontend + session persistence + streaming

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Methodology

What I Built

- Built a domain-specific agriculture corpus
- Implemented RAG pipeline
  - ingestion
  - chunking
  - embeddings
  - Qdrant indexing
  - retrieval + metadata filtering
  - grounded answer synthesis
- Designed specialist routingcrop
  - pest
  - market
  - irrigation via Agent B
  - soil via MCP
- Added guardrails, timeouts, sessions, and graceful fallback

04

Results

Results

Configuration	Retrieval Metrics	Generation Metrics	Interpretation
Baseline localdet top-5, 900/180	P@5 = 0.15 R@5 = 0.60 MRR = 0.4517	Faithfulness = 0.55 Correctness = 0.431 Relevance = 0.4667	Functional but noisy retrieval. The system often finds at least one relevant source, but ranking quality remains weak and answer quality is uneven.
Baseline localdet top-3, 700/120	P@3 = 0.2166 R@3 = 0.65 MRR = 0.5092	Faithfulness = 0.55 Correctness = 0.502 Relevance = 0.5164	Best-performing configuration. Smaller chunks and lower top-k reduced retrieval noise and improved overall answer quality.

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Challenges

Challenges

Agent B failure

Cause: SQLite schema mismatch
Fix: reset persisted DB / align schema

Production reliability

Fix: fallback logic, timeouts, session persistence, guardrails

Qdrant evaluation script failure

Cause: API mismatch (search vs query_points)
Fix: updated retrieval evaluation code

Weak retrieval on some queries

Fix: compared configurations and selected better chunking / top-k

Failure Cases

Olive irrigation query
weak crop-specific irrigation retrieval

Early blight query
disease term underweighted vs crop term

Abstract harvest-vs-hold query
retrieval distracted by loosely related documents

Limitations

- heuristic generation evaluator
- lightweight reranking
- deterministic embedding mode prioritizes reproducibility
- Agent B fallback kept for reliability

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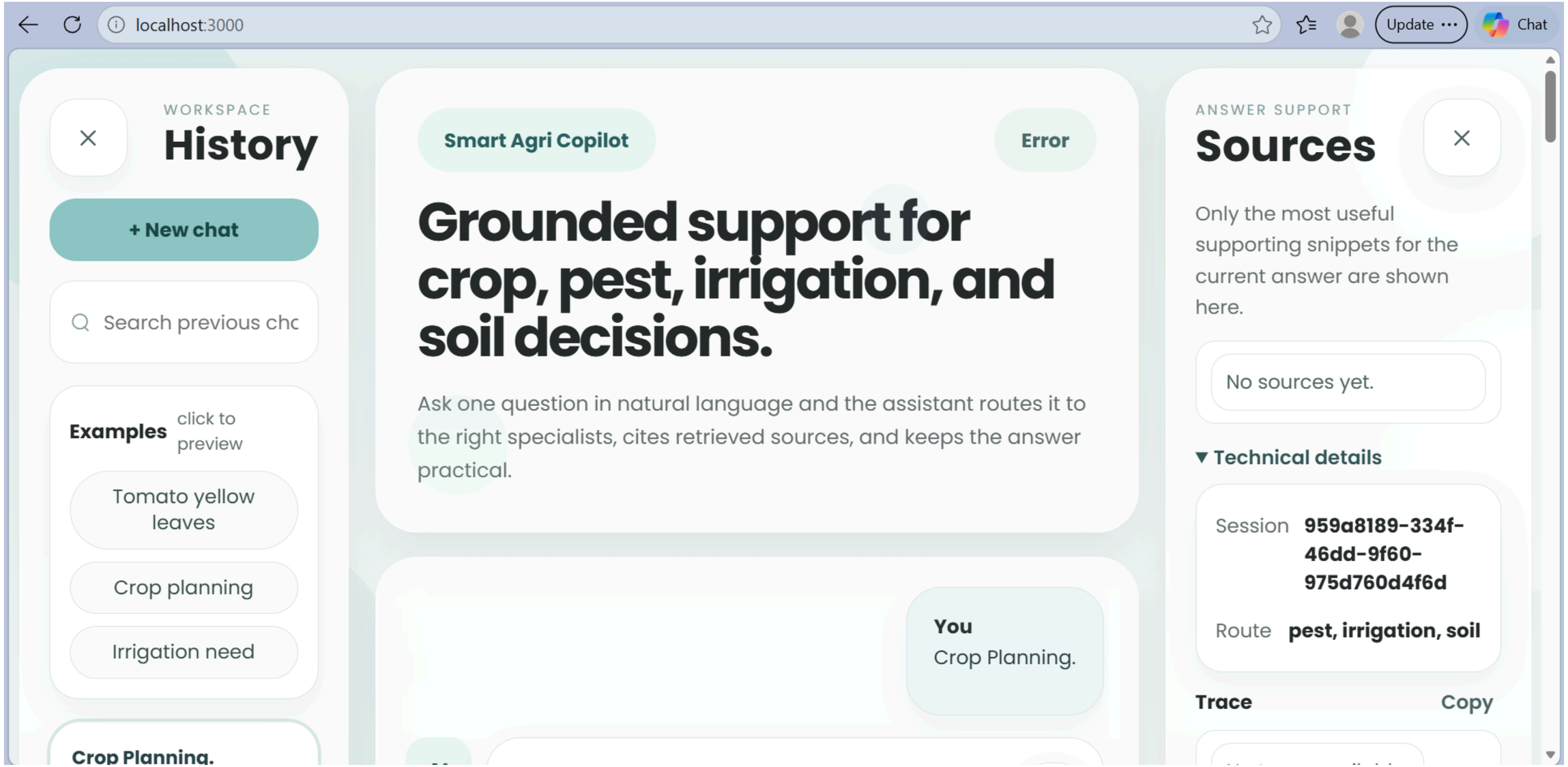
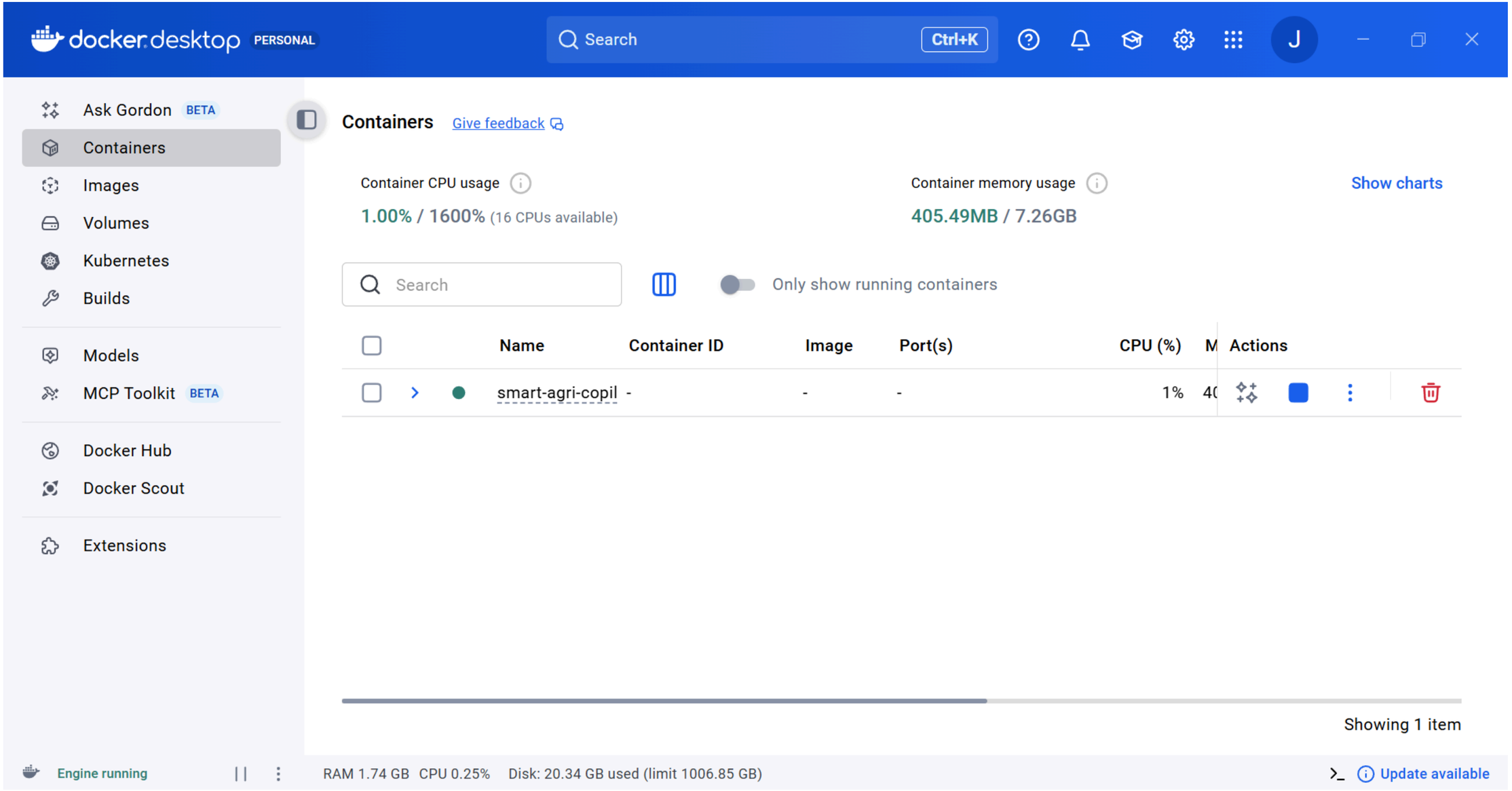
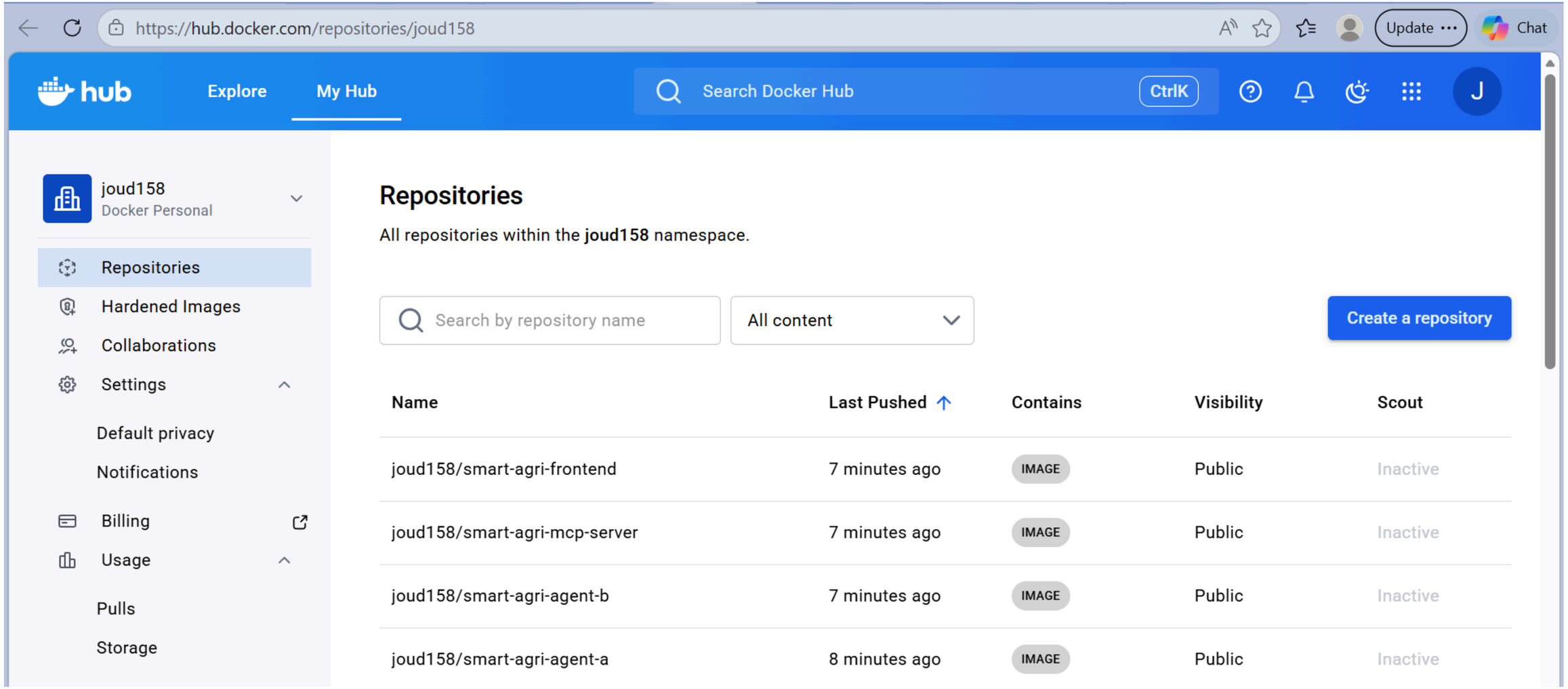
Demo


localhost:3000
Update
Chat
WORKSPACE
History
+ New chat
Search previous chc
Examples
click to preview
Tomato yellow leaves
Crop planning
Irrigation need
Crop Planning.
Smart Agri Copilot
Error
Grounded support for crop, pest, irrigation, and soil decisions.
Ask one question in natural language and the assistant routes it to the right specialists, cites retrieved sources, and keeps the answer practical.
You
Crop Planning.
ANSWER SUPPORT
Sources
Only the most useful supporting snippets for the current answer are shown here.
No sources yet.
Technical details
Session 959a8189-334f-46dd-9f60-975d760d4f6d
Route pest, irrigation, soil
Trace
Copy


docker desktop PERSONAL
Search
Ctrl+K
J
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Containers
Give feedback
Container CPU usage
1.00% / 1600% (16 CPUs available)
Container memory usage
405.49MB / 7.26GB
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Actions
smart-agri-copil
-
-
-
1%
Showing 1 item
Engine running
RAM 1.74 GB CPU 0.25% Disk: 20.34 GB used (limit 1006.85 GB)
Update available

- CI/CD pipeline was implemented using DockerHub workflows



Thank **you!**